AI-DRIVEN E-WASTE RECOVERY USING HYDROGEN DECREPITATION

SORTING PERFECTLY ALIGNS WITH THE WASTE STREAM WITH BESPOKE EXTRACTIONS, UNLOCKING PURE STRATEGIC METALS.

E-WASTE AT A GLANCE

- **62 MILLION TONNES** Generated globally annually
- Fastest growing waste stream
- Rich source of critical metals
- Low recycling efficiency (<20%)
- Urban mining for a sustainable future

COMMON E-WASTE SOURCES



STRATEGIC METALS PRESENT

METAL		APPLICATION
Li	Lithium	EV Batteries
Co	Cobalt	Battery Cathodes
Ni	Nickel	High-energy Cells
Cu	Copper	Electrical Wiring
Au	Gold	PCB Contacts
Ag	Silver	Conductive Layers
Pd	Palladium	Electronic Components
Nd	Neodymium	Permanent Magnets

THE CORE TECHNOLOGY

HYDROGEN DECREPITATION



CHEMICALS USED

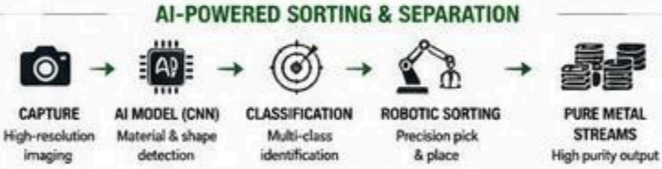
- **HYDROGEN (H_2)** Creates lattice expansion and enables decrepitation
- **NITROGEN (N_2)** Provides inert atmosphere
- **ARGON (Ar)** Prevents oxidation of reactive metals
- **ETHANOL (C_2H_5OH)** Cleaning & degreasing of materials
- **WATER (H_2O)** Washing & impurity removal
- **CITRIC ACID ($C_6H_8O_7$)** Green leaching agent (eco-friendly)
- **SULFURIC ACID (H_2SO_4)** Industrial leaching for metal recovery
- **HYDROCHLORIC ACID (HCl)** Metal dissolution & purification
- **HYDROGEN PEROXIDE (H_2O_2)** Oxidizing agent for impurity removal

HYDROGEN DECREPITATION PROCESS (HOW IT WORKS)

1. Hydrogen enters the cracked battery materials.
2. Hydrogen diffuses into the metal lattice.
3. Lattice expands due to hydrogen absorption.
4. Micro-cracks form and weaken the structure.
5. Active material separates naturally from the current collector.
6. Fine powder is collected for efficient downstream recovery.

> "AI-POWERED SORTING PERFECTLY ALIGNS WITH HETEROGENEOUS E-WASTE STREAMS, ENABLING BESPOKE EXTRACTION PATHWAYS THAT UNLOCK HIGH-PURITY STRATEGIC METALS WHILE MINIMIZING MATERIAL LOSS."

AI-POWERED SORTING & SEPARATION



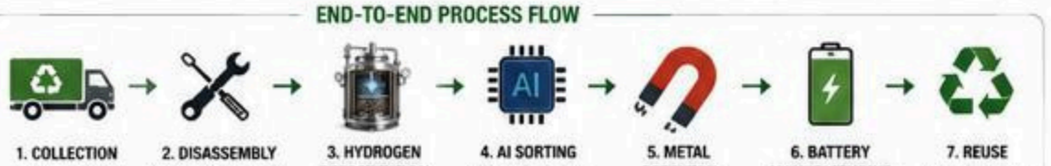
KEY BENEFITS

- **HIGHER PURITY** Up to battery-grade metals
- **LOWER CHEMICAL CONSUMPTION** Green & efficient process
- **FASTER RECOVERY** Reduced time & energy
- **LOWER EMISSIONS** Environmentally sustainable
- **CIRCULAR ECONOMY** Close the loop of materials
- **BATTERY-GRADE METALS** For renewable energy future

CIRCULAR ECONOMY LOOP



END-TO-END PROCESS FLOW



OUR VISION

Combining Hydrogen Decrepitation with AI-driven sorting to unlock the true value of e-waste and secure tomorrow's critical metals sustainably.

SUPPORTS SDGs

7 AFFORDABLE AND CLEAN ENERGY · 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE · 12 RESPONSIBLE CONSUMPTION AND PRODUCTION · 13 CLIMATE ACTION

DATA SOURCES

- United Nations University (UNU)
- Global E-waste Monitor 2024
- CPCB, MoEFCC, NITI Aayog Reports
- Industry & Research Publications